FYUGP B.Sc. PHYSICS HONOURS

ELECTRODYNAMICS II

Semester V · Core in Major

MODULE 1: ELECTRIC AND MAGNETIC FIELDS IN MATTER

Model questions on topics the past papers miss

Section A: 1 mark each

Polarization

1 Write the relation between the induced dipole moment $\mathbf{p}$ of an atom and the field $\mathbf{E}$. Give the polarizability of a model atom of radius a. **(1)**

2 Write the bound volume charge density ρ_b in terms of the polarization $\mathbf{P}$. **(1)**

3 Write the field inside a uniformly polarized sphere of radius R. **(1)**

Magnetization

4 Write the torque on a magnetic dipole $\mathbf{m}$ in a uniform field $\mathbf{B}$. **(1)**

5 Write the bound surface current $\mathbf{K}_b$ in terms of the magnetization $\mathbf{M}$ and the normal $\hat{\mathbf{n}}$. **(1)**

6 Write $\mathbf{B}$ in terms of $\mathbf{H}$ for a linear magnetic medium of susceptibility χ_m. **(1)**

Section B: 2 marks each, answer all

Polarization and bound charge

7 Explain physically why a string of head-to-tail dipoles leaves net charge at its two ends. **(2)**

8 State the boundary condition on $D^{\perp}$ and on $\mathbf{D}^{\parallel}$ at an interface, in terms of free charge and polarization. **(2)**

Magnetization and bound currents

9 Explain physically why a uniformly magnetized slab carries a surface current, although each atom carries only a tiny loop current. **(2)**

10 Show that $\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$ and explain why $\mathbf{H}$ cannot be found from the free current alone in a bar magnet. **(2)**

11 Explain why a magnetic dipole in a uniform field feels a torque but no net force, and why this changes in a non-uniform field. **(2)**

Section C: 4 marks each, answer any two

Polarization

12 A point charge $Q = 1\,\mu$C is fixed at the origin. A dipole $p = 1$ nC m is free to ride on a circle of radius $R = 0.1$ m about the charge, with its axis tangent to the circle. Using $\mathbf{F} = (\mathbf{p} \cdot \nabla)\mathbf{E}$, show that the force on the dipole is $\mathbf{F} = \dfrac{Q\,\mathbf{p}}{4\pi\varepsilon_0 R^3}$ and find its magnitude. **(4)**

13 A sphere of radius $R = 0.1$ m carries polarization $\mathbf{P} = k\mathbf{r}$, with $k = 2 \times 10^{-6}$ C/m^2. Find the bound volume charge ρ_b, the bound surface charge σ_b, and the total bound charge. **(4)**

14 A solid sphere of radius 2 cm is uniformly polarized with $P = 1.0 \times 10^{-6}$ C/m^2. Find the field inside the sphere and the dipole moment of the sphere. Take $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m. **(4)**

15 A dielectric sphere of relative permittivity $\varepsilon_r = 3$ is placed in a uniform field $E_0 = 1000$ V/m. Find the field inside the sphere and the polarization inside it. **(4)**

Magnetization

16 A uniformly magnetized sphere of radius 5 cm has $M = 1.0 \times 10^4$ A/m. Find the magnetic field inside the sphere and its magnetic dipole moment. Take $\mu_0 = 4\pi \times 10^{-7}$ H/m. **(4)**

17 A long solenoid with $n = 1000$ turns/m carries a current $I = 2$ A and is filled with a linear material of $\chi_m = 2 \times 10^{-5}$. Find $\mathbf{H}$, $\mathbf{B}$ inside, and the bound surface current density on the surface of the material. **(4)**

Section D: 10 marks each, answer any one

Polarization

18 Starting from the bound charges $\rho_b = -\nabla \cdot \mathbf{P}$ and $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ of a uniformly polarized sphere, derive the field inside the sphere and the potential outside it. Show that the outside field is that of a dipole $\mathbf{p} = \frac{4}{3}\pi R^3 \mathbf{P}$. **(10)**

19 A conducting sphere of radius a carries a charge Q and is surrounded by linear dielectric of permittivity ε out to radius b. Find $\mathbf{D}$, $\mathbf{E}$ and the potential at the centre of the sphere, relative to infinity. **(10)**

Magnetization

20 Explain diamagnetism using the orbital motion of an electron in a magnetic field. Derive the change in the orbital dipole moment $\Delta\mathbf{m} = -\dfrac{e^2 R^2}{4m_e}\mathbf{B}$, and explain why it is opposite to $\mathbf{B}$ and weaker than paramagnetism. **(10)**